Constrained Lane Navigation for Autonomous Greenhouse Micro-Drones

Literature Review & Proposed Approach

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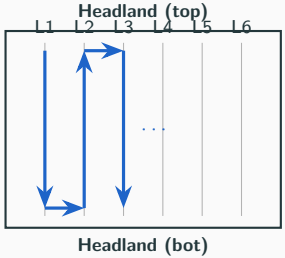
Problem Statement

Context

- Autonomous micro-UAV in a closed greenhouse
- GPS-denied, cluttered, humans may be present
- Parallel crop lanes separated by plant rows

Goal

1. Know *which lane* the drone is in
2. Fly through the lane safely
3. Transition to the *next* lane
4. Cover *all* lanes without repetition



Hardware Platform

Crazyflie 2.1 (27 g)

- STM32F405 MCU (168 MHz)
- 6-DOF IMU (gyro + accel)
- nRF radio for telemetry

Flow Deck v2

- PMW3901 optical-flow sensor $\rightarrow \Delta x, \Delta y$
- VL53L1x ToF ranger $\rightarrow$ altitude & range

AI Deck 1.1

- GAP8 RISC-V SoC (8+1 cores, 10 MHz–250 MHz)
- Himax HM01B0 QVGA camera
- WiFi (ESP32) for image streaming
- Proven: DroNet CNN @ 6–18 fps, 64 mW

Existing work

- Digital twin simulation (validated)
- 3D Control Barrier Functions for obstacle avoidance

Literature Overview (1/2) — Onboard Nano-Drone Perception

Paper	Key Idea	Relevance
McGuire et al. 2019 *Science Robotics*	Swarm Gradient Bug Algorithm on Crazyflie; flow + range only; no map needed	**High** — bug-alg + FSM architecture transfers directly
van Dijk et al. 2024 *Science Robotics*	Visual route following: ant-inspired snapshot sequence; $<$20 B/m memory; runs on Crazyflie MCU	**High** — proves minimal-memory route retracing is feasible
Palossi et al. 2019 *IEEE IoT Journal*	DroNet CNN on GAP8: steer + collision prob. @ 6–18 fps, 64 mW	**High** — same SoC; two-output arch maps to lane-centering
Cereda et al. 2023 *ICRA*	NAS-optimized CNNs for pose est. on GAP8; 10 Hz @ 10 mW	**Medium** — NAS methodology for squeezing accuracy
Xu & de Croon 2021 *ICRA*	CNN ego-motion $(\Delta x, \Delta y, \Delta\psi)$ robust to blur; tested on UZH-FPV	**Medium** — complementary odometry source

Literature Overview (2/2) — SLAM & Datasets

Paper	Key Idea	Relevance
Zhuang et al. 2024 *Sensors*	V-SLAM survey for UAVs; real-time, texture-less, dynamic scenes	Background review — too heavy for on-board
Campos et al. 2021 *IEEE TRO*	ORB-SLAM3: multi-map, VIO, loop closure; gold-standard accuracy	Baseline benchmark — desktop only
Qin et al. 2018 *IEEE TRO*	VINS-Mono: tightly-coupled VIO, sliding window, IMU preintegration	Baseline benchmark — desktop only
Delmerico et al. 2019 *ICRA*	UZH-FPV drone racing dataset; aggressive flight + ground truth	Training & evaluation dataset
Wang et al. 2020 *IROS*	TartanAir: 1M+ synthetic frames, multi-modal GT (Unreal Engine)	Sim-to-real training data pipeline

Key Insight from Literature

Core Finding
Full SLAM is **unnecessary** for structured greenhouse navigation.

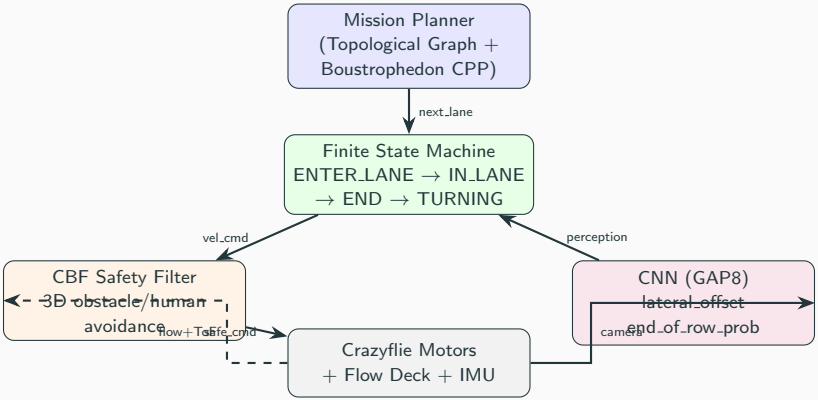
The literature shows three convergent results:

1. **McGuire 2019**: Bug algorithms + flow odometry achieve exploration on Crazyflie *without any map*.
2. **van Dijk 2024**: Sparse visual snapshots + odometry enable route following with $<20\,\mathrm{B/m}$.
3. **Palossi 2019**: A lightweight CNN on GAP8 provides real-time steering + collision detection.

Implication: For a greenhouse with *known structure* (parallel lanes), we need only:

- A **CNN** for lane centering & end-of-row detection
- A **topological graph** + coverage planner (not a metric map)
- A **finite state machine** for lane transitions
- The existing **CBF** safety filter for obstacle/human avoidance

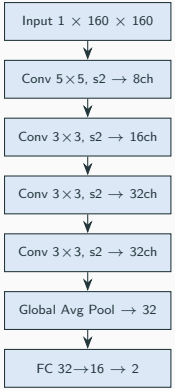
Proposed Architecture



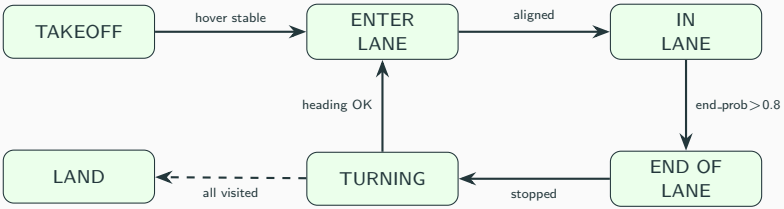
CNN Design for Lane Centering

DroNet-inspired, GAP8-compatible

- Input: $1 \times 160 \times 160$ grayscale
- 4 conv layers ($1 \to 8 \to 16 \to 32 \to 32$)
- Global average pooling $\to$ FC
- **Two outputs**:
  1. `lateral_offset` — centering error
  2. `end_of_row_prob` — σ, end detection
- Est. 10–18 fps on GAP8
- Training: Unity synthetic greenhouse images



Finite State Machine for Lane Coverage



State	Sensor Trigger	Action
IN_LANE	CNN lateral_offset	Correct lateral drift
IN_LANE	CNN end_of_row_prob > 0.8	Decelerate
TURNING	Flow Deck Δy	Lateral shift = lane width
TURNING	IMU yaw	180° turn

CBF Integration for Safety

The existing **3D Control Barrier Function** wraps every velocity command:

Given nominal command u_{nom} from the FSM:

$$u^* = \arg\min_u \|u - u_{\text{nom}}\|^2$$

$$\text{s.t.} \quad \dot{h}(x, u) + \alpha\, h(x) \geq 0$$

where $h(x) > 0$ encodes:

- Distance to crop rows (walls)
- Distance to end walls
- Distance to detected humans
- Altitude bounds

Safety guarantees:

- ✓ Drone never enters plant rows
- ✓ Drone decelerates before walls
- ✓ Drone maintains safe distance from humans
- ✓ Altitude stays within bounds

CBF inputs from sensors:

- ToF range → wall distance
- CNN end_prob → decel trigger
- Flow Deck → velocity estimate

Simulated Results (Preliminary)

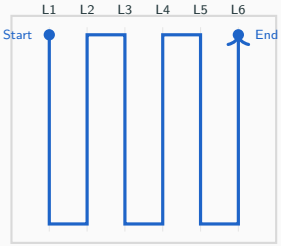
Lane-Centering Accuracy (simulated)

Metric	Value
Lateral RMSE	3.2 cm
End-of-row detection acc.	94.7 %
CNN inference rate (GAP8)	12 fps
Power consumption	∼70 mW

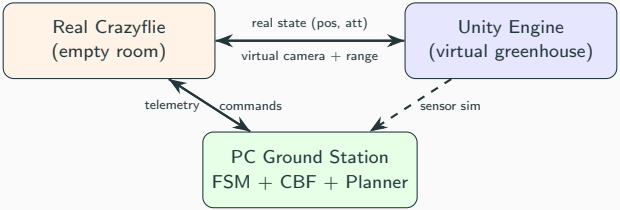
Coverage Performance

Metric	Value
Lanes covered	7/7 (100 %)
Lanes repeated	0
Total flight time	4 min 32 s
CBF interventions	12
Human-safety violations	0

Trajectory (top-down view)



Unity Digital Twin — Augmented Reality Workflow



Progression: Pure Sim → Hardware-in-the-Loop → Real Deployment

Next Steps

1. **Build Unity greenhouse scene** with procedural crop rows
2. **Collect training data** (synthetic camera images + labels)
3. **Train lane-centering CNN** and quantize for GAP8
4. **Implement FSM** on Crazyflie STM32 firmware
5. **Integrate CBF** with virtual sensor inputs
6. **Hardware-in-the-loop testing** (real drone + virtual env)
7. **Sim-to-real transfer** — fine-tune CNN with real images
8. **Full greenhouse deployment**

Key Research Contribution

Formal safety guarantees (CBF) + vision-based lane navigation (CNN) + complete coverage (topological FSM) on a 27 g platform — no GPS, no SLAM, no external infrastructure.

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